

Android 7+ Par Self-Signed Certificate Install Kaise Karein

Android 7 (Nougat) se Google ne ek bada security change kiya tha. Iske baad se Android apps automatically user-installed certificates ko trust nahi karti. Iska matlab hai ke agar aap Burp Suite ya Charles ka self-signed certificate install karte hain, toh apps usko ignore kar dengi aur HTTPS traffic intercept nahi ho payegi .

Pehle Android 6 aur us se neeche ke versions mein user certificate install karna kaafi aasan tha, lekin Android 7 ke baad yeh trick kaam nahi karti .

Android 7+ Mein Kya Change Hua Hai?

Android 7 se system ne user-added CA certificates ko trust karna band kar diya. Pehle jo certificates hum install karte the, woh user certificate store mein jaate the. Android 7+ par apps sirf system certificate store ko trust karti hain. User store mein install certificate sirf browser traffic ke kaam aata hai, apps usko ignore kar deti hain .

Android 14 mein yeh aur strict ho gaya hai. System certificates ab APEX container mein move kar diye gaye

hain jo read-only mount point hai. Isko directly modify nahi kiya ja sakta .

☒ **Tareeqa 1: Rooted Device Par System Certificate Install Karna (Best Method)**

Agar aapka device rooted hai toh yeh sab se reliable tareeqa hai. Root ke saath aap system certificate store mein Burp/Charles certificate daal sakte hain.

Step 1: Burp Suite Se Certificate Export Karein

Burp Suite mein **Proxy** → **Options** par jayein. **Import / export CA certificate** mein **Certificate in DER format** select karein aur burp.der file save karein .

Step 2: Certificate Ko PEM Format Mein Convert Karein

Android ko PEM format chahiye. OpenSSL use karein:

```
openssl x509 -inform der -in burp.der -out  
burp.pem
```

Step 3: Certificate Ka Hash Generate Karein

Android system certificate ko specific naming convention chahiye. Hash generate karein:

```
openssl x509 -inform PEM -subject_hash_old -in  
burp.pem | head -1
```

Output kuch is tarah aayega: 9a5ba575 (Burp Suite ka hash hai).

Step 4: Certificate Rename Karein

Certificate ko hash ke naam se rename karein aur .0 extension daalein:

```
mv burp.pem 9a5ba575.0
```

Step 5: Certificate Ko System Directory Mein Push Karein

ADB ke zariye certificate ko system directory mein daalein:

```
adb root  
adb remount  
adb push 9a5ba575.0 /system/etc/security/cacerts/  
adb shell chmod 644  
/system/etc/security/cacerts/9a5ba575.0
```

Step 6: Device Reboot Karein

```
adb reboot
```

Step 7: Verify Karein

Settings → Security → Trusted Credentials → System mein jaake Portswigger CA check karein .

Tareeqa 2: Magisk Module Use Karein (Easy & Safe)

Agar Magisk se root kiya hai toh yeh sab se aasan aur safe tareeqa hai. NCC Group ka **ConscriptTrustUserCerts** module use karein .

Steps:

1. Burp certificate ko user store mein install karein (normal tareeqe se)
2. **ConscriptTrustUserCerts.zip** module download karein
3. Magisk app → Modules → Install from storage → ZIP select karein
4. Reboot karein
5. Module automatically user certificate ko system store mein copy kar dega har boot par

Yeh method safe hai kyunke yeh system partition ko permanently modify nahi karta.

Tareeqa 3: Emulator Use Karein (No Root Required)

Agar physical device root nahi karna chahti toh Android emulator use kar sakti hain. MEmu, MuMu, ya Android Studio emulator root mode support karte hain .

MEmu Example:

1. MEmu System Settings mein **Root mode** ON karein
2. Disk sharing mode "Independent system" karein (write permission ke liye)
3. Certificate ko system directory mein push karein:

```
adb root
adb shell "mount -o rw,remount /system"
adb push 9a5ba575.0 /system/etc/security/cacerts/
adb shell "chmod 644
/system/etc/security/cacerts/9a5ba575.0"
```

Android 14+ Special Note

Android 14 mein system certificate store /apex/com.android.conscrypt/cacerts mein move ho gaya hai jo read-only mount point hai. Isko directly modify nahi kiya ja sakta . Isliye Android 14+ par **Magisk module (ConscryptTrustUserCerts)** hi best option hai .

✗ Tareeqa 4: App Ke Network Security Config Modify Karna (Root Nahin)

Agar root nahi hai toh app ke AndroidManifest.xml mein networkSecurityConfig add kar sakte hain .

Step 1: res/xml/network_security_config.xml banayein:

```
<network-security-config>
    <debug-overrides>
        <trust-anchors>
            <certificates src="user" />
        </trust-anchors>
    </debug-overrides>
</network-security-config>
```

Step 2: Manifest mein add karein:

```
<application
android:networkSecurityConfig="@xml/network_security_config">
```

Lekin yeh method sirf un apps par kaam karega jinmein aap code change kar sakte hain.



Summary – Kaunsa Tareeqa Choose Karein?

Rooted Physical Device – Manual system certificate install ya Magisk module use karein .

Rooted Device with Magisk – ConscriptTrustUserCerts module best hai .

Emulator – Root mode enable karein aur system certificate install karein .

App Source Code Available – Network security config modify karein .

⚠ **Important:** Rooting se device warranty void ho sakti hai. Yeh process sirf educational aur authorized penetration testing ke liye hai. Kisi doosre ke network par bina permission ke traffic intercept karna applicable cyber laws ke under punishable hai .

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